



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LEARNER GUIDE

For

Certified Lean Six Sigma Yellow Belt (CLSSYB)

Course Code: C524

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1	1 June 2026	Initial release — CLSSYB Learner Guide covering the 10 hands-on labs.	Dr. Alfred Ang
2	15 June 2026	Refreshed labs against the CSSC Yellow Belt reference.	Dr. Alfred Ang
3	1 July 2026	Aligned lab content with the Contoso Service Desk case-study scenario.	Dr. Alfred Ang
4	23 September 2026	Major content expansion: guide restructured to follow the DMAIC roadmap end to end; Lean Six Sigma concepts imported from the reference trainer deck and the CSSC body of knowledge; 4 elective labs added (Affinity/Kano, VSM/Takt, Solution Matrix/Benchmarking/ FMEA, Descriptive Statistics/Implementation Plan) and Lab 4 (PDCA charter) reclassified as elective; interactive problem-solving tools (5 Whys, Fishbone, collaborative Pareto, NovaSPC) integrated into the Analyze labs; added quick-reference formulas, the eight wastes and an expanded glossary.	Dr. Alfred Ang
5	23 September 2026	Integrated the SIPOC & Process Map Builder (alfredang.github.io/sipoc/) into Lab 3 and elective Lab 12, with a tool walkthrough added to the Define phase and the online toolkit slide updated to list all five tools.	Dr. Alfred Ang
6	23 September 2026	Fixed clipped headings: slide titles now auto-fit to a single line so long lab titles	Dr. Alfred Ang

		can no longer overprint the LAB/ELECTIVE chips beneath them.	
7	20 July 2026	Table of contents restored to a live, updatable Word field (previously flattened to static text), so it refreshes on open or F9 while still rendering correctly in the distributed PDF.	Dr. Alfred Ang
9	23 September 2026	Labs reissued as ACTIVITY PACKS in the Tertiary Infotech house format: one folder per activity named "NN - Title", each holding a Facilitator Guide, a Learner Worksheet and a Checklist (DOCX + PDF) alongside its own mock data (data/*.csv) and blank worksheets (templates/*.csv). The previous labs/ Markdown layout is superseded and archived.	Dr. Alfred Ang

TABLE OF CONTENTS

Introduction.....	5
Course Learning Outcomes.....	5
Before You Start.....	6
FOUNDATIONS — Six Sigma Foundations (15%).....	7
Activity 1 — Yellow Belt Role, Certification Paths, and Improvement Scenario [Core].....	8
DEFINE — Define — Scope the Problem (25%).....	9
Activity 2 — Lean, Six Sigma, Waste, Voice of Customer, and Value [Core].....	9
Activity 3 — SIPOC, Process Mapping, Handoffs, and SME Support [Core].....	10
Activity 4 — PDCA Small Improvement Project Charter [Elective].....	11
Activity 5 — DMAIC Overview, Problem Statement, Scope, and Stakeholders [Core].....	12
Activity 11 — Affinity Diagram and Kano Analysis [Elective].....	13
MEASURE — Measure — Quantify Performance (25%).....	14
Activity 6 — Data Collection, KPIs, Check Sheets, and Basic Metrics [Core].....	14
Activity 12 — Value Stream Map and Takt Time [Elective].....	15
ANALYZE — Analyze — Find the Root Cause (20%).....	16
Activity 7 — Pareto, Run Charts, Variation, Yield, DPU, and DPMO [Core].....	16
Activity 8 — Root Cause Analysis with 5 Whys, Fishbone, and Evidence [Core].....	18
IMPROVE — Improve — Fix the Cause (10%).....	19
Activity 9 — Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen [Core].....	19
Activity 13 — Solution Selection Matrix, Benchmarking, and FMEA [Elective].....	20
CONTROL — Control — Hold the Gain (5%).....	21
Activity 10 — Control Plan, A3 Summary, Handover, and Certification Readiness [Core].....	21
Activity 14 — Descriptive Statistics and Implementation Planning [Elective].....	22
Quick Reference — Formulas You Should Know.....	23
Quick Reference — The Eight Wastes (DOWNTIME).....	24
Consolidating What You Learned.....	24
Glossary.....	25

Introduction

This Learner Guide accompanies the course Certified Lean Six Sigma Yellow Belt (CLSSYB) (C524), conducted by Tertiary Infotech Academy Pte Ltd. It follows the DMAIC roadmap end to end — Define, Measure, Analyze, Improve, Control — and provides step-by-step instructions for every hands-on activity. Core activities are assessed; elective activities are provided as additional practice and are run when time allows.

The course content is grounded in the body of knowledge published by The Council for Six Sigma Certification (CSSC) in 'Six Sigma: A Complete Step-by-Step Guide', so what you learn here matches the recognised Yellow Belt standard.

Every activity uses one continuous scenario — the Contoso Service Desk, where IT tickets take too long to be assigned and employees must chase for status. By the end of the course your lab outputs form a complete improvement package: role definition, VOC/CTQ, process maps, data collection plan, analysis, root cause, countermeasures and a control plan.

Course Learning Outcomes

- LO1: Define a project and establish the scope of work using Six Sigma's Define, Measure and Analyse phases.
- LO2: Apply Lean and Six Sigma concepts — value, waste, defects and variation — to a work process.
- LO3: Map a process using SIPOC, process maps and value stream maps to expose handoffs and waste.
- LO4: Collect and analyse process data using check sheets, Pareto charts, run charts and basic metrics.
- LO5: Identify root causes using 5 Whys, Fishbone analysis and evidence-based prioritisation.
- LO6: Recommend improvement and control actions to sustain gains, and continue your Lean Six Sigma development.

Before You Start

What you need

- A laptop with a spreadsheet application (Microsoft Excel, Google Sheets or LibreOffice Calc).
- A browser, for the interactive problem-solving tools used in the Analyze labs.
- The course slides and this Learner Guide, downloaded from <https://lms-tms.tertiaryinfotech.com>.
- A work process of your own to think about — the tools apply far better when the example is real.

The interactive problem-solving toolkit

Five browser-based tools are used during the labs. No installation, licence or sign-up is required, and nothing you enter leaves your browser.

- SIPOC & Process Map Builder — a guided SIPOC that enforces the 5-7 step rule, tags pain points, and generates the swimlane and handoff table from your actor assignments: <https://alfredang.github.io/sipoc/>
- 5 Whys — build and share a 5 Whys chain: <https://alfredang.github.io/5whys/>
- Fishbone Diagram — build an Ishikawa cause-and-effect diagram: <https://alfredang.github.io/fishbone/>
- Pareto Chart (collaborative) — your team brainstorms and votes in one live session and the Pareto chart builds itself: <https://alfredang.github.io/paretochart/>
- NovaSPC — run charts, SPC charts and process capability from your own CSV: <https://alfredang.github.io/novaspc/>

Core and elective activities

- Core activities are completed by everyone and map directly to the DMAIC phases.
- Elective labs extend the same scenario with additional Lean Six Sigma tools; complete them if time allows or as post-course practice.
- All labs build on the same Contoso Service Desk scenario, so outputs carry forward from one lab to the next.

The Contoso Service Desk data set

Every activity works from one real-shaped data set: two weeks of service desk activity — 400 tickets, of which 96 contained at least one defect, with 120 defects recorded across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal.

- Yield 76% · DPU 0.30 · DPO 0.0500 · DPMO 50,000 · sigma level ~3.1.
- Pareto vital few: Delayed assignment (40%) and Missing information (25%) — 65% of all defects.
- The run chart hides a genuine special cause: performance shifts from day 7, when the ITSM platform was migrated.
- Each lab folder carries its own data/ (mock data to analyse) and templates/ (worksheets to complete).

Where to find each activity's files

Each activity is a self-contained folder under activities/ — activities/NN - <Name>/ — holding a Facilitator Guide, a Learner Worksheet and a Checklist (each as DOCX and PDF), plus data/ (the CSV mock data) and templates/ (the blank CSV worksheets you fill in). Open the CSV files in Excel, Google Sheets or LibreOffice Calc, and always work on a copy.

Conventions used in every activity

- Each lab states its objective, the deliverable you produce, the steps, and a check to confirm you are done.
- Tables shown in the steps can be built in a spreadsheet or on the worksheet provided.
- Where a lab uses an online tool, the tool URL is shown with the step.
- Keep every lab output — they combine into your final improvement package and are your revision material.

FOUNDATIONS — Six Sigma Foundations (15%)

Quality · Lean · Six Sigma · Lean Six Sigma · Belt roles · The DMAIC roadmap

Key concepts

- What is Quality — Quality is conformance to customer requirements — not merely defect-free output.
- What is Lean — A method to maximise customer value by systematically removing waste and improving flow.
- What is Six Sigma — A data-driven method to reduce variation and defects — targeting 3.4 defects per million.
- Lean Six Sigma — The combined method: faster flow AND fewer defects, driven by data.
- Belt roles — White, Yellow, Green, Black and Master Black Belt — who does what on a project.
- The DMAIC roadmap — Define, Measure, Analyze, Improve, Control — the disciplined improvement path.

Activity 1 — Yellow Belt Role, Certification Paths, and Improvement Scenario [Core]

Objective: Explain the Yellow Belt role and select a suitable improvement scenario (A1).

Goal: Establish what a Lean Six Sigma Yellow Belt is accountable for, how the belt pathway progresses from White to Master Black Belt, and choose a small, manageable improvement scenario to carry through the whole course.

What you'll build

A Yellow Belt responsibility table and a selected improvement scenario. (Tools and techniques: Belt role matrix, CSSC certification pathway, project selection criteria.)

Step-by-step

1. Define the Yellow Belt role — list at least four responsibilities and your contribution to each.

2. Compare the White, Yellow, Green, Black and Master Black Belt pathways in a table.
3. Review the Contoso Service Desk scenario: tickets are slow to be assigned and status is unclear.
4. Test your scenario against the 'good project' criteria — day-to-day work, manageable, aligned to business goals, data available.
5. Record why a Yellow Belt supports rather than leads this improvement.

Data and worksheets for this activity

- data/service-desk-tickets.csv — 400 rows of mock data (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)
- templates/belt-pathway-comparison.csv — worksheet to complete (belt, typical_role, training_days, leads_or_supports, typical_project_scope)
- templates/project-selection-criteria.csv — worksheet to complete (criterion, does_the_scenario_meet_it, evidence)
- templates/yellow-belt-responsibilities.csv — worksheet to complete (responsibility, your_contribution, belt_that_leads_it)

The scenario brief and the two weeks of ticket data are in data/.

Check your work

You can state the Yellow Belt role in one sentence and justify your scenario against all four selection criteria.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/01 - Yellow Belt Role, Certification Paths, and Improvement Scenario/.

DEFINE — Define — Scope the Problem (25%)

VOC · CTQ · Project charter · Problem statement · Scope · SIPOC · Process mapping

Key concepts

- Voice of the Customer — Capture customer needs in their own words before deciding anything.
- Critical to Quality — Translate each VOC need into a specific, measurable CTQ requirement.
- Project charter — Problem statement, goal, scope, team and benefit — the project's contract.
- Problem statement — Process, time period, measurable issue and impact — and never a solution.
- SIPOC — The macro as-is map: Suppliers, Inputs, Process, Outputs, Customers.

- Process mapping — Flowcharts and swimlanes expose the handoffs where delay and defects are born.

Activity 2 — Lean, Six Sigma, Waste, Voice of Customer, and Value [Core]

Objective: Apply Lean and Six Sigma concepts — value, waste, defects, variation (K1, A2).

Goal: Separate what the customer actually values from the waste that surrounds it. Capture the Voice of the Customer, translate it into measurable CTQ requirements, and run a waste walk using the eight wastes (DOWNTIME).

What you'll build

A VOC-to-CTQ translation table, a value-added analysis, and a waste walk log. (Tools and techniques: VOC, CTQ tree, DOWNTIME eight wastes, value-added analysis.)

Step-by-step

6. Capture the Voice of the Customer — record at least five verbatim customer statements.
7. Translate each VOC statement into a need, then into a measurable CTQ requirement.
8. Classify each process activity as value-added, business-value-added or non-value-added.
9. Conduct a waste walk and tag each observation against the eight wastes (DOWNTIME).
10. Distinguish a defect (output fails CTQ) from waste (effort the customer will not pay for).
11. Identify which single waste type appears most often in your scenario.

Data and worksheets for this activity

- data/voice-of-customer.csv — 10 rows of mock data (id, customer_statement, source, date)
- data/waste-walk-observations.csv — 10 rows of mock data (obs_id, observation, process_step, time_lost_min, waste_type)
- templates/value-added-analysis.csv — worksheet to complete (step_no, activity, VA_BVA_NVA, justification)
- templates/voc-to-ctq.csv — worksheet to complete (voc_id, customer_statement, need, ctq_requirement, measure, target)

Tag every waste-walk observation with one of the eight DOWNTIME waste types.

Check your work

Every CTQ is measurable with a target, and each waste observation is tagged to one of the eight waste types.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/02 - Lean, Six Sigma, Waste, Voice of Customer, and Value/.

Activity 3 — SIPOC, Process Mapping, Handoffs, and SME Support [Core]

Objective: Map a process with SIPOC and a detailed process map to expose handoffs (A2, A3).

Goal: Build the macro 'as-is' view with SIPOC, then drill into a detailed process map that shows every actor, system and handoff — the points where delay and defects are actually created.

What you'll build

A completed SIPOC and a detailed process map with pain points marked. (Tools and techniques: SIPOC & Process Map Builder, process flowchart, swimlane map, standard process symbols.)

Step-by-step

12. Set the process boundaries — agree the explicit start and stop points first.
13. Build the SIPOC with the online builder: Suppliers, Inputs, Process (5-7 high-level steps), Outputs, Customers.

<https://alfredang.github.io/sipoc/>

14. Expand into a detailed process map: Step, Actor, Activity, System, Handoff.
15. Assign an actor to each step — the tool generates the swimlane and detects every handoff for you.
16. Tag pain points on the steps — waiting, rework loop, unclear ownership, duplicate entry or missing decision rule (at least three).
17. Prepare SME notes for the Green Belt: what you observed and what needs validation.

Data and worksheets for this activity

- data/process-steps.csv — 7 rows of mock data (step_no, activity, actor, system, process_time_min, wait_time_min, handoff_to)
- templates/handoff-register.csv — worksheet to complete (handoff_no, from_actor, to_actor, trigger, owner_both_sides, pain_point)
- templates/sipoc.csv — worksheet to complete (suppliers, inputs, process_step, outputs, customers)

data/process-steps.csv holds the as-is step, actor, system and timing data.

Check your work

Run 'Check my SIPOC' in the tool — all five columns populated, 5-7 steps, at least three pain points, and every handoff owned on both sides.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/03 - SIPOC, Process Mapping, Handoffs, and SME Support/.

Activity 4 — PDCA Small Improvement Project Charter [Elective]

Objective: Charter a small improvement using PDCA with a measurable goal (A1, A2).

Goal: Not every improvement needs a full DMAIC project. Use PDCA to charter a small, fast improvement that a Yellow Belt can run under guidance — with a problem statement, a measurable goal and a defined check point.

What you'll build

A one-page PDCA charter with problem statement, goal, scope and success measure. (Tools and techniques: PDCA cycle, problem/goal statements, scope in-out, stakeholder list.)

Step-by-step

18. Map the four PDCA phases to concrete actions for your scenario.
19. Write the problem statement: process, time period, measurable issue and impact — no solution.
20. Write a measurable goal statement: metric, baseline, target and date.
21. Define scope with an explicit in-scope / out-of-scope table to prevent scope creep.
22. Identify stakeholders and the decision you will make after the Check phase.
23. Confirm the improvement is small enough to test within two weeks.

Data and worksheets for this activity

- templates/pdca-charter.csv — worksheet to complete (section, content)

Baseline figures for the goal statement come from data/daily-summary.csv.

Check your work

Your goal statement contains a metric, a baseline, a target and a date, and your problem statement names no solution.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/04 - PDCA Small Improvement Project Charter/.

Activity 5 — DMAIC Overview, Problem Statement, Scope, and Stakeholders [Core]

Objective: Support the Define phase of a DMAIC project (A1, A2).

Goal: Step up from PDCA to the full DMAIC roadmap. Clarify what each phase delivers, where a Yellow Belt contributes most, and prepare defensible Define-phase information for the project leader.

What you'll build

A DMAIC phase table, refined problem statement, stakeholder map and benefit estimate. (Tools and techniques: DMAIC roadmap, project charter, stakeholder analysis, CTQ linkage.)

Step-by-step

24. Summarise all five DMAIC phases: purpose, key deliverable and Yellow Belt support role.
25. Refine the problem statement so it is specific, measurable and solution-free.
26. Identify stakeholders and classify each by influence and interest.
27. Define the project scope and state the expected benefit in business terms.
28. Link the problem back to the CTQ requirements captured in Activity 2.
29. Confirm which DMAIC phases a Yellow Belt can support most strongly.

Data and worksheets for this activity

- templates/dmaic-phase-table.csv — worksheet to complete (phase, purpose, key_deliverable, yellow_belt_support_role)
- templates/stakeholder-map.csv — worksheet to complete (stakeholder, interest, influence_H_M_L, interest_H_M_L, engagement_approach)

Check your work

Your problem statement passes the 'no solution named' test and every stakeholder has a defined engagement approach.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/05 - DMAIC Overview, Problem Statement, Scope, and Stakeholders/.

Activity 11 — Affinity Diagram and Kano Analysis [Elective]

Objective: Organise customer requirements and classify them by satisfaction impact (K2).

Goal: Two Define-phase tools from the original course. Use an Affinity Diagram to cluster unstructured VOC input into themes, then apply Kano Analysis to classify requirements as Must-Be, One-Dimensional or Delighter.

What you'll build

An Affinity Diagram of clustered VOC themes and a Kano classification table. (Tools and techniques: Affinity Diagram, Kano Analysis, VOC clustering.)

Step-by-step

30. Write each VOC statement on a separate note — one idea per note.
31. Cluster related notes into natural groups without talking, then name each group.
32. Classify each requirement as Must-Be, One-Dimensional or Delighter.
33. Plot the requirements on the Kano diagram and identify where to invest first.

Data and worksheets for this activity

- data/voice-of-customer.csv — 10 rows of mock data (id, customer_statement, source, date)
- templates/affinity-groups.csv — worksheet to complete (voc_id, statement, affinity_group)
- templates/kano-classification.csv — worksheet to complete (requirement, kano_class_MustBe_OneDim_Delighter, why, invest_first)

Check your work

Every VOC note sits in exactly one named affinity group and carries a Kano classification.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/11 - Affinity Diagram and Kano Analysis/.

MEASURE — Measure — Quantify Performance (25%)

The 8 wastes · Data types · Data collection plan · Check sheets · Yield · DPMO · Sigma level

Key concepts

- The 8 wastes — DOWNTIME — Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.
- Types of data — Continuous vs discrete; nominal vs ordinal — the data type drives the tool choice.
- Data collection plan — What, who, when and how — with operational definitions to remove ambiguity.
- Check sheets — A simple structured form is the most reliable way to capture process data.

- Process metrics — Yield, DPU, DPO and DPMO quantify how well the process actually performs.
- Sigma level — Convert DPMO into a sigma level to benchmark the process against Six Sigma.

Activity 6 — Data Collection, KPIs, Check Sheets, and Basic Metrics [Core]

Objective: Plan and execute data collection against defined quality standards (A4).

Goal: Replace anecdote with evidence. Define KPIs with unambiguous operational definitions, design a check sheet, plan the sampling approach, and understand which data type you are collecting — because the data type drives the tool.

What you'll build

A data collection plan, an operational definition set and a working check sheet. (Tools and techniques: KPI definition, operational definitions, check sheets, sampling, data types.)

Step-by-step

34. Classify your data: continuous vs discrete, nominal vs ordinal — and why it matters.
35. Define KPIs with operational definitions, data source and collection frequency.
36. Design a check sheet capturing date, ticket ID, type, assignment time and defect category.
37. Define mutually exclusive defect categories so every observation lands in exactly one.
38. Plan sampling: how many, how often, by whom — and identify possible bias.
39. State which KPI best reflects the customer pain point from your VOC work.

Data and worksheets for this activity

- data/daily-summary.csv — 10 rows of mock data (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- data/service-desk-tickets.csv — 400 rows of mock data (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)
- templates/check-sheet.csv — worksheet to complete (date, ticket_id, ticket_type, assignment_time_min, defect_category, rework_required, notes)
- templates/data-collection-plan.csv — worksheet to complete (kpi, operational_definition, data_type, data_source, frequency, who_collects, bias_risk)

data/service-desk-tickets.csv is the full two-week extract (400 tickets).

Check your work

Two different people reading your operational definition would record the same value for the same event.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/06 - Data Collection, KPIs, Check Sheets, and Basic Metrics/.

Activity 12 — Value Stream Map and Takt Time [Elective]

Objective: Quantify flow, lead time and takt time across the value stream (A3, A4).

Goal: Extend process mapping into Lean metrics. Map the value stream with process and wait times, calculate process cycle efficiency, then compute takt time to see whether the process can meet customer demand.

What you'll build

A value stream map with lead time, process time and a calculated takt time. (Tools and techniques: SIPOC & Process Map Builder, value stream mapping, lead time, WIP, takt time.)

Step-by-step

40. Map the value stream: each step with its process time and the wait time between steps. Start from your Activity 3 process map.

<https://alfredang.github.io/sipoc/>

41. Total the value-added time and the lead time, then compute process cycle efficiency.

42. Calculate takt time = available working time / customer demand.

43. Compare cycle time against takt time to identify the bottleneck step.

Data and worksheets for this activity

- data/process-steps.csv — 7 rows of mock data (step_no, activity, actor, system, process_time_min, wait_time_min, handoff_to)
- templates/takt-time.csv — worksheet to complete (input, value)
- templates/value-stream-map.csv — worksheet to complete (step_no, activity, process_time_min, wait_time_min, VA_or_NVA)

Check your work

Your lead time equals the sum of all process and wait times, and takt time is expressed per unit.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/12 - Value Stream Map and Takt Time/.

ANALYZE — Analyze — Find the Root Cause (20%)

Variation · Pareto · Run charts · 5 Whys · Fishbone · Multi-voting · Evidence

Key concepts

- Variation — Common cause is built into the process; special cause is an external, assignable signal.
- Pareto analysis — The 80/20 rule separates the vital few causes from the trivial many.
- Run charts — Plot the metric over time to reveal trend, shift, cluster and oscillation.
- 5 Whys — Ask why repeatedly to drill from the symptom down to an actionable cause.
- Fishbone diagram — Organise candidate causes by category — Manpower, Method, Machine, Material, Measurement.
- Evidence testing — A cause is only a root cause when the data you collected supports it.

Activity 7 — Pareto, Run Charts, Variation, Yield, DPU, and DPMO [Core]

Objective: Analyse process performance data to quantify and prioritise (A3, A4).

Goal: Turn collected data into decisions. Apply the Pareto principle to find the vital few categories, use a run chart to see behaviour over time, and calculate the standard Six Sigma process metrics.

What you'll build

A Pareto chart, a run chart and calculated yield, DPU, DPO, DPMO and sigma level. (Tools and techniques: Pareto Chart tool (collaborative), NovaSPC, yield/DPU/DPO/DPMO, sigma level, variation.)

Step-by-step

44. Build the Pareto table: category, count, percentage and cumulative percentage.
45. In your group, open the collaborative Pareto tool — one member creates the session and shares the code, then everyone joins, brainstorms and votes to produce the live Pareto chart.

<https://alfredang.github.io/paretochart/>

46. Identify the vital few categories driving about 80% of the problem.
47. Export your assignment-time data as CSV and plot a run chart in NovaSPC.

<https://alfredang.github.io/novaspc/>

48. Interpret the run chart for trend, shift, cluster and oscillation patterns.
49. Calculate yield, DPU, DPO and DPMO from your defect data.
50. Convert DPMO to a sigma level and interpret what it says about the process.

51. Distinguish common-cause from special-cause variation and why the response differs.

Data and worksheets for this activity

- data/assignment-times.csv — 400 rows of mock data (date, ticket_id, assignment_time_min)
- data/daily-summary.csv — 10 rows of mock data (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- data/defect-counts.csv — 6 rows of mock data (defect_category, count)
- data/service-desk-tickets.csv — 400 rows of mock data (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)
- templates/pareto-table.csv — worksheet to complete (defect_category, count, percent, cumulative_percent)
- templates/process-metrics.csv — worksheet to complete (metric, formula, your_calculation)

Upload data/assignment-times.csv to NovaSPC for the run chart. Worked answers are in solution/pareto-table-answers.csv and solution/process-metrics-answers.csv — attempt the lab before opening them.

Check your work

Your cumulative percentage column reaches 100%, and you can state the sigma level with the DPMO it came from.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/07 - Pareto, Run Charts, Variation, Yield, DPU, and DPMO/.

Activity 8 — Root Cause Analysis with 5 Whys, Fishbone, and Evidence [Core]

Objective: Identify root causes of variation using structured analysis (A3).

Goal: Move from symptom to cause. Apply the 5 Whys to drill past the obvious, organise candidate causes on a Fishbone diagram, then test each candidate against the evidence you actually collected.

What you'll build

A completed 5 Whys chain, a Fishbone diagram and an evidence-tested cause shortlist. (Tools and techniques: 5 Whys tool, Fishbone tool, 5M categories, brainstorming, multi-voting.)

Step-by-step

52. Write the problem statement precisely — the effect you are explaining.

53. Complete a 5 Whys chain with the online 5 Whys tool, continuing until you reach an actionable cause.

<https://alfredang.github.io/5whys/>

54. Build a Fishbone diagram with the online Fishbone tool using the 5M categories: Manpower, Method, Machine, Material, Measurement.

<https://alfredang.github.io/fishbone/>

55. Brainstorm candidate causes into each category — no evaluation during generation.

56. Use multi-voting to shortlist the most likely causes as a team.

57. Test each shortlisted cause against your Activity 7 data — does the evidence support it?

58. State why the team must not jump straight to solutions.

Data and worksheets for this activity

- data/daily-summary.csv — 10 rows of mock data (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- data/defect-counts.csv — 6 rows of mock data (defect_category, count)
- data/root-cause-evidence.csv — 8 rows of mock data (evidence_id, observation, source, supports_cause)
- templates/cause-evidence-test.csv — worksheet to complete (candidate_cause, evidence_that_supports_it, evidence_that_contradicts_it, verdict)
- templates/fishbone-causes.csv — worksheet to complete (category, possible_cause, supported_by_evidence)
- templates/five-whys.csv — worksheet to complete (why_level, question, answer, evidence_needed, evidence_found)

Test every candidate cause against the Activity 7 data before shortlisting it.

Check your work

Each shortlisted root cause is supported by named evidence, and your 5 Whys chain ends at something you can act on.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/08 - Root Cause Analysis with 5 Whys, Fishbone, and Evidence/.

IMPROVE — Improve — Fix the Cause (10%)

Countermeasures · 5S · Poka-Yoke · Standard work · Kaizen · Solution selection · Piloting

Key concepts

- Generating solutions — Brainstorm widely against the proven root cause before evaluating anything.
- Solution selection — Score candidate solutions against weighted criteria before committing.
- 5S — Sort, Set in order, Shine, Standardise, Sustain — for physical and digital work.
- Poka-Yoke — Mistake proofing makes the error difficult or impossible to make in the first place.
- Standard work — Document the improved method so anyone can repeat it the same way.
- Piloting — Test the change at small scale to expose issues before full rollout.

Activity 9 — Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen [Core]

Objective: Recommend improvement actions that address the proven root cause (A5).

Goal: Generate and select countermeasures that attack the root cause you proved — not the symptom. Apply the core Lean improvement tools: 5S, poka-yoke, visual management, standard work and Kaizen.

What you'll build

A prioritised countermeasure set with a 5S plan, a poka-yoke design and standard work. (Tools and techniques: Brainstorming, 5S, poka-yoke, visual management, standard work, Kaizen.)

Step-by-step

59. Generate countermeasures for each proven root cause — quantity before quality.
60. Prioritise using an impact vs effort matrix to find the quick wins.
61. Apply 5S thinking — Sort, Set in order, Shine, Standardise, Sustain — including to digital work.
62. Design one poka-yoke that makes the error difficult or impossible to make.
63. Draft standard work so the improved method is repeatable by anyone.
64. Plan a Kaizen event or pilot to test the countermeasure at small scale first.
65. Distinguish a containment countermeasure from a permanent solution.

Data and worksheets for this activity

- templates/5s-plan.csv — worksheet to complete (S, what_it_means_here, action, owner)
- templates/countermeasures.csv — worksheet to complete (root_cause, countermeasure, type_5S_pokayoke_standardwork, impact_H_M_L, effort_H_M_L, expected_effect)

Check your work

Every countermeasure traces back to a proven root cause, and your poka-yoke prevents rather than detects the error.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/09 - Countermeasures, 5S, Mistake Proofing, Standard Work, and Kaizen/.

Activity 13 — Solution Selection Matrix, Benchmarking, and FMEA [Elective]

Objective: Evaluate and de-risk candidate solutions before implementation (A5, K2).

Goal: Three Improve-phase tools from the original course. Score solutions against weighted criteria, benchmark against outstanding practice, and use FMEA to find how a proposed change could fail before it does.

What you'll build

A scored solution selection matrix, a benchmarking summary and an FMEA with RPN. (Tools and techniques: Solution selection matrix, benchmarking, FMEA, RPN scoring.)

Step-by-step

66. Agree the selection criteria and weight each by importance.
67. Score every candidate solution against the weighted criteria and rank the results.
68. Benchmark your process against an internal or external reference and note the gap.
69. Build an FMEA: failure mode, effect, cause, then score Severity, Occurrence and Detection.
70. Calculate $RPN = S \times O \times D$ and address the highest-RPN failure modes first.

Data and worksheets for this activity

- data/candidate-solutions.csv — 5 rows of mock data (solution_id, candidate_solution, est_cost, est_effort_days)
- templates/fmea.csv — worksheet to complete (process_step, failure_mode, effect, cause, severity_1_10, occurrence_1_10, detection_1_10, RPN, action)
- templates/solution-selection-matrix.csv — worksheet to complete (solution, impact_w5, cost_w3, ease_w2, weighted_score, rank)

Check your work

Your matrix ranks solutions by weighted score, and every FMEA row has an RPN and an action for the highest scores.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/13 - Solution Selection Matrix, Benchmarking, and FMEA/.

CONTROL — Control — Hold the Gain (5%)

Control plan · Visual management · SOPs · Huddles · A3 · Handover · Certification

Key concepts

- Control plan — Metric, target, monitoring method, frequency, owner and reaction plan.
- Process control — A process is in control when it varies consistently within expected limits.
- Visual management — Boards and dashboards keep the improved performance visible to everyone.
- Standard operating procedures — Written instructions that lock in the improved method.
- Team huddles — Short, regular stand-ups that surface problems while they are still small.

Activity 10 — Control Plan, A3 Summary, Handover, and Certification Readiness [Core]

Objective: Recommend control actions to sustain process performance (A5).

Goal: Hold the gain. Build a control plan that names the metric, the target, the monitoring frequency, the owner and the reaction plan — then tell the whole improvement story on a single A3 page and hand it over.

What you'll build

A control plan, an A3 one-page summary, a handover checklist and a readiness plan. (Tools and techniques: Control plan, SPC thinking, visual management, A3 report, handover.)

Step-by-step

71. Create the control plan: metric, target, monitoring method, frequency, owner, reaction plan.
72. Define the visual management approach — board, dashboard or huddle — that keeps it visible.
73. Specify the reaction plan: exactly what happens when a metric falls outside target.
74. Write the A3 summary: background, current state, goal, analysis, countermeasures, results, follow-up.
75. Prepare the handover checklist so the process owner can sustain it without you.
76. Complete your personal certification readiness plan — which topics need most review.

Data and worksheets for this activity

- data/assignment-times.csv — 400 rows of mock data (date, ticket_id, assignment_time_min)
- templates/a3-summary.csv — worksheet to complete (a3_section, content)
- templates/control-plan.csv — worksheet to complete (process_step, metric, target, owner, check_frequency, response_plan)
- templates/handover-checklist.csv — worksheet to complete (handover_item, accepted_by, date, notes)

Re-plot data/assignment-times.csv in NovaSPC to show the post-improvement position.

Check your work

Every control plan row has a named owner and a reaction plan, and your A3 fits on one page.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/10 - Control Plan, A3 Summary, Handover, and Certification Readiness/.

Activity 14 — Descriptive Statistics and Implementation Planning [Elective]

Objective: Summarise data numerically and plan the rollout (A4, A5).

Goal: Close the loop. Compute the measures of central tendency and dispersion that describe your data, then convert the selected countermeasures into a dated implementation plan with owners and barriers identified.

What you'll build

A descriptive statistics summary and a dated implementation plan. (Tools and techniques: Mean, median, range, standard deviation, implementation planning.)

Step-by-step

77. Compute the mean and median for your assignment-time data and compare them.
78. Compute the range and standard deviation to describe the spread.
79. Explain what the mean-versus-median difference reveals about outliers and skew.
80. Write the implementation plan: action, owner, date, barriers and mitigation.

Data and worksheets for this activity

- data/assignment-times.csv — 400 rows of mock data (date, ticket_id, assignment_time_min)
- data/daily-summary.csv — 10 rows of mock data (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- data/service-desk-tickets.csv — 400 rows of mock data (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)

- templates/descriptive-statistics.csv — worksheet to complete (statistic, formula_or_function, your_value)
- templates/implementation-plan.csv — worksheet to complete (action, owner, start_date, due_date, barrier, mitigation)

data/assignment-times.csv is the raw column for the statistics.

Check your work

You can explain why the mean and median differ in your data, and every implementation action has an owner and a date.

Note: The Learner Worksheet, Checklist, mock data and blank templates for this activity are in activities/14 - Descriptive Statistics and Implementation Planning/.

Quick Reference — Formulas You Should Know

Process performance metrics

- Yield = (Good units / Total units) x 100
- DPU (Defects Per Unit) = Defects / Units
- DPO (Defects Per Opportunity) = Defects / (Units x Opportunities per unit)
- DPMO (Defects Per Million Opportunities) = DPO x 1,000,000
- First Pass Yield (FPY) = units passing with no rework / total units started
- Rolled Throughput Yield (RTY) = FPY(step 1) x FPY(step 2) x ... x FPY(step n)
- Process Cycle Efficiency = Value-added time / Total lead time
- Takt time = Available working time / Customer demand

Sigma level reference

- 1 sigma — 690,000 DPMO (about 31% yield)
- 2 sigma — 308,000 DPMO (about 69% yield)
- 3 sigma — 66,800 DPMO (about 93.3% yield)
- 4 sigma — 6,210 DPMO (about 99.38% yield)
- 5 sigma — 233 DPMO (about 99.977% yield)
- 6 sigma — 3.4 DPMO (about 99.99966% yield)

Quick Reference — The Eight Wastes (DOWNTIME)

- D — Defects: output that fails the requirement and must be corrected or redone.
- O — Overproduction: producing more, or earlier, than the customer needs.
- W — Waiting: work or people idle, waiting for the next step, an approval or information.
- N — Non-utilised talent: skills and ideas of people not being used.

- T — Transport: unnecessary movement of materials, work items or information between places.
- I — Inventory: work in progress, backlogs and queues sitting between steps.
- M — Motion: unnecessary movement of people, or switching between systems and screens.
- E — Extra-processing: doing more work to the output than the customer requires or values.

Consolidating What You Learned

- Keep these slides, this Learner Guide and your activity outputs — they are your working reference.
- Re-read your own activity outputs; they follow the same Contoso Service Desk scenario end to end.
- Be ready to define Lean, Six Sigma and Lean Six Sigma, and explain how they differ.
- Be ready to name the eight wastes and give a service-industry example of each.
- Be ready to explain each DMAIC phase, what it delivers and which tools belong to it.
- Be ready to calculate yield, DPU, DPO and DPMO from raw data and read off the sigma level.
- Be ready to explain how the Fishbone diagram and 5 Whys are used together to find a root cause.
- Re-work the activities from memory — being able to produce the tools unaided is the best practice.
- Full attendance across both days gives you the complete DMAIC picture.

Glossary

- **Lean** — A method to maximise customer value by systematically identifying and removing waste.
- **Six Sigma** — A data-driven method to reduce variation and defects; the target is 3.4 defects per million opportunities.
- **Lean Six Sigma** — The combined method — Lean improves speed and flow, Six Sigma improves consistency and accuracy.
- **DMAIC** — Define, Measure, Analyze, Improve, Control — the Six Sigma improvement roadmap.
- **PDCA** — Plan, Do, Check, Act — a lighter improvement cycle used for small, fast improvements.
- **VOC** — Voice of the Customer — customer needs and expectations expressed in the customer's own words.
- **CTQ** — Critical to Quality — a specific, measurable requirement translated from a VOC statement.
- **SIPOC** — Suppliers, Inputs, Process, Outputs, Customers — the macro 'as-is' process map.

- **Muda** — The Japanese term for waste — any activity that consumes resource but creates no customer value.
- **DOWNTIME** — Mnemonic for the eight wastes: Defects, Overproduction, Waiting, Non-utilised talent, Transport, Inventory, Motion, Extra-processing.
- **Value-added (VA)** — An activity the customer would be willing to pay for because it changes the product or service.
- **Non-value-added (NVA)** — An activity that consumes resource but adds nothing the customer values — pure waste.
- **Defect** — An output that fails to meet the CTQ requirement.
- **Common cause variation** — Natural, always-present variation built into the process; addressed by changing the process.
- **Special cause variation** — Unusual variation traceable to a specific assignable event; addressed by investigating that event.
- **Pareto principle** — The 80/20 rule — roughly 80% of effects come from 20% of causes.
- **Run chart** — A plot of a metric over time, used to spot trend, shift, cluster and oscillation patterns.
- **5 Whys** — Repeatedly asking 'why' to drill from a symptom down to an actionable root cause.
- **Fishbone (Ishikawa) diagram** — A cause-and-effect diagram organising candidate causes into categories such as the 5Ms.
- **5M categories** — Manpower, Method, Machine, Material, Measurement — standard Fishbone categories.
- **Multi-voting** — A team technique to narrow a long list of candidate causes to an agreed shortlist.
- **DPU / DPO / DPMO** — Defects per unit / per opportunity / per million opportunities — standard defect-rate metrics.
- **Yield** — The percentage of output produced without defects.
- **RTY** — Rolled Throughput Yield — the probability a unit passes through every process step with no rework.
- **Sigma level** — A common yardstick for process performance, derived from DPMO.
- **COPQ** — Cost of Poor Quality — the internal failure, external failure, appraisal and prevention costs of defects.
- **5S** — Sort, Set in order, Shine, Standardise, Sustain — a Lean method for organising the workplace.
- **Poka-Yoke** — Mistake proofing — designing the process so an error is difficult or impossible to make.
- **Standard work** — The documented current best-known method for performing a task.
- **Kaizen** — Continuous improvement through many small changes, made by everyone.
- **Takt time** — Available working time divided by customer demand — the pace the process must run to meet demand.

- **Control plan** — The document naming the metric, target, monitoring method, frequency, owner and reaction plan.
- **Specification limits (LSL/USL)** — The acceptable limits set by the customer; outside them is a defect.
- **Control limits (LCL/UCL)** — Limits calculated from the process itself, typically the mean plus/minus three standard deviations.
- **SPC** — Statistical Process Control — monitoring a process over time to predict and prevent problems.
- **Gemba** — 'The real place' — going to where the work actually happens to observe the process directly.